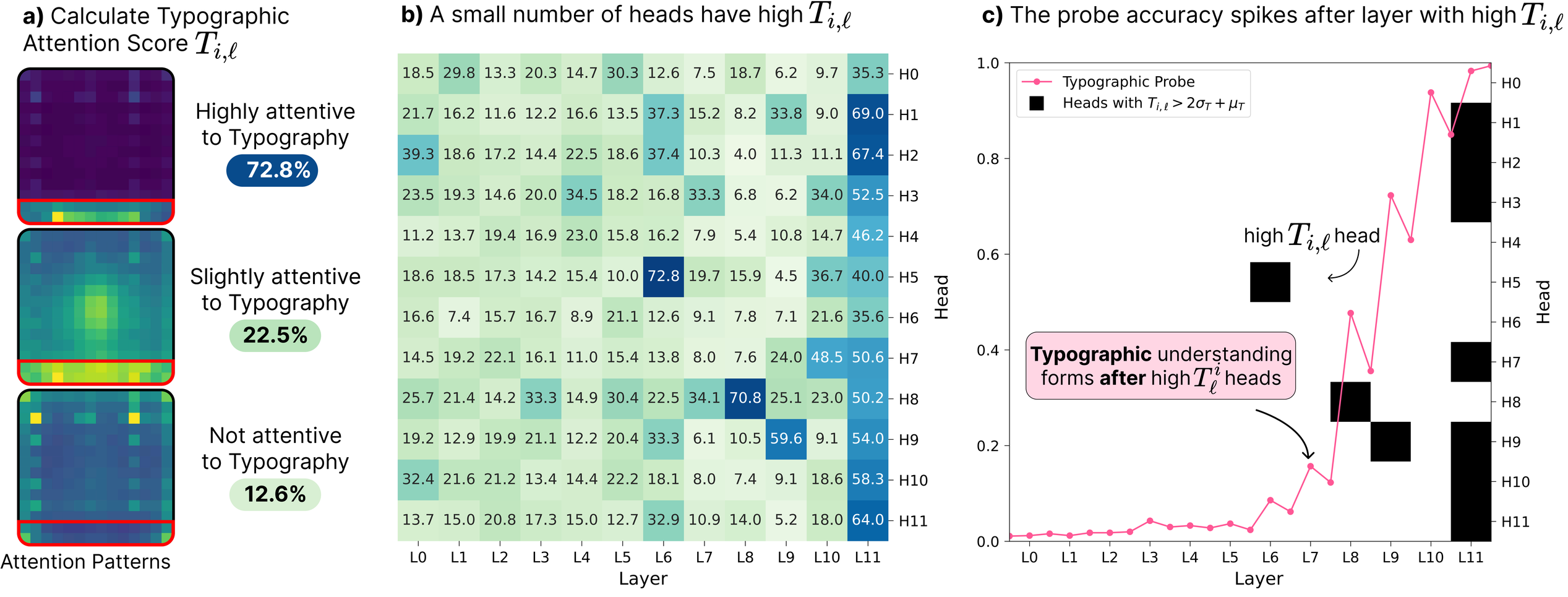
a) Calculate Typographic Attention Score $T_{i,\ell}$
Highly attentive to Typography
72.8%
Slightly attentive to Typography
22.5%
Not attentive to Typography
12.6%
Attention Patterns
b) A small number of heads have high $T_{i,\ell}$
18.5 29.8 13.3 20.3 14.7 30.3 12.6 7.5 18.7 6.2 9.7 35.3 H0
21.7 16.2 11.6 12.2 16.6 13.5 37.3 15.2 8.2 33.8 9.0 69.0 H1
39.3 18.6 17.2 14.4 22.5 18.6 37.4 10.3 4.0 11.3 11.1 67.4 H2
23.5 19.3 14.6 20.0 34.5 18.2 16.8 33.3 6.8 6.2 34.0 52.5 H3
11.2 13.7 19.4 16.9 23.0 15.8 16.2 7.9 5.4 10.8 14.7 46.2 H4
18.6 18.5 17.3 14.2 15.4 10.0 72.8 19.7 15.9 4.5 36.7 40.0 H5
16.6 7.4 15.7 16.7 8.9 21.1 12.6 9.1 7.8 7.1 21.6 35.6 H6
14.5 19.2 22.1 16.1 11.0 15.4 13.8 8.0 7.6 24.0 48.5 50.6 H7
25.7 21.4 14.2 33.3 14.9 30.4 22.5 34.1 70.8 25.1 23.0 50.2 H8
19.2 12.9 19.9 21.1 12.2 20.4 33.3 6.1 10.5 59.6 9.1 54.0 H9
32.4 21.6 21.2 13.4 14.4 22.2 18.1 8.0 7.4 9.1 18.6 58.3 H10
13.7 15.0 20.8 17.3 15.0 12.7 32.9 10.9 14.0 5.2 18.0 64.0 H11
L0 L1 L2 L3 L4 L5 L6 L7 L8 L9 L10 L11
Layer
Head
c) The probe accuracy spikes after layer with high $T_{i,\ell}$
Typographic Probe
Heads with $T_{i,\ell} > 2\sigma_T + \mu_T$
high $T_{i,\ell}$ head
Typographic understanding forms after high T_ℓ^i heads
1.0
0.8
0.6
0.4
0.2
0.0
L0 L1 L2 L3 L4 L5 L6 L7 L8 L9 L10 L11
Layer
H0 H1 H2 H3 H4 H5 H6 H7 H8 H9 H10 H11
Head